Frequency and Coverage Damage under LoRA Adaptation: A Matched Four-Condition Decomposition on BRACS and TCGA-UT

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Abstract

Class imbalance changes two things at once: how strongly each class influences the training loss (frequency) and which rare-class patches are available (coverage). An earlier decomposition separated these two factors for a frozen encoder with a logistic-regression readout. The mitigation studies instead used a Virchow2 encoder with low-rank adaptation (LoRA), so frozen-encoder shares need not describe their recovery. This experiment repeats the decomposition with the LoRA recipe of the mitigation studies on BRACS and TCGA-UT at an imbalance ratio of 100. Four matched conditions train fresh adapters and heads from the same initialization: a balanced reference, frequency only, coverage only, and both. Fixed loss weights change class influence without changing which patches are sampled. The contrasts estimate total damage, frequency-only and coverage-only damage, their interaction, and the recovery from restoring balanced class influence under this recipe. The recovered share provides a conditional benchmark for rebalancing methods. On BRACS, training on the imbalanced patches with balanced class influence costs 3.09 of the 5.07 lost percentage points. Restoring balanced class influence recovers 1.97 points, a share of 0.39 [0.24, 0.56]. On TCGA-UT, both single-factor damage intervals span zero; the interaction accounts for 1.40 of the 1.58 lost points. Restoring balanced class influence yields a share consistent with full recovery (1.01 [0.86, 1.22]). Logit adjustment recovered similar amounts in the mitigation studies, descriptively consistent with the coverage-only residual on BRACS.

1 Motivation

On TCGA-UT and BRACS, an imbalance ratio of $\rho = 100$ lowered the patient-macro balanced accuracy of a frozen Virchow2 encoder (Zimmermann et al., 2024) with a logistic-regression readout by 2.91 and 7.65 percentage points (Queisler, 2026b,a). Those studies decomposed the damage into frequency-only damage from skewed class influence in the loss, coverage-only damage from thinner patch coverage of rare classes, and their interaction. Frequency-only damage accounted for about two-fifths of the TCGA-UT damage and three-fifths of the BRACS damage.

The mitigation studies then trained a low-rank adaptation (LoRA) update (Hu et al., 2022) of the same encoder together with a linear head (Queisler, 2026d,c,e). Several methods that act on class influence, such as logit adjustment, recovered about two-fifths of the damage on BRACS and nearly all of it on TCGA-UT. The frozen-encoder shares do not establish whether these recoveries approach what restoring balanced class influence could achieve. LoRA changes the encoder during training, so coverage and frequency may act differently. This experiment therefore asks: *under the LoRA recipe of the mitigation studies, how much of the damage at* $\rho = 100$ *comes from frequency, from coverage, and from their interaction, and how much does restoring balanced class influence recover?*

The question follows from the observed mitigation results. The answer provides a descriptive benchmark for those results, rather than a prediction made before they were known.

2 Design

The design fixes the data and training recipe of the mitigation studies and varies only two factors: the training patch allocation and each class's share of the loss.

Data. Both datasets use the three patient-disjoint splits and the ten allocation draws per split of the mitigation studies, giving 30 blocks per dataset. BRACS (Brancati et al., 2022) has seven classes; each block trains on 10 patients per class and 2,240 patches. TCGA-UT (Komura et al., 2022) has 30 classes; each block trains on 20 patients per class and 19,200 patches. Within a block, the balanced allocation $r1$ gives every class the same number of patches, and the imbalanced allocation $r100$ gives the classes exponentially decreasing patch counts with a ratio of 100 between the largest and smallest class. Both allocations reuse the patients, patch order, and class-to-rank assignment of the mitigation studies. Validation and test patients, class order, preprocessing, and the evaluation metric are unchanged.

Conditions. Table 1 defines the four conditions. The frequency-only condition F trains on the balanced patches but weights the loss so that each class has the influence it would have under $r100$. The coverage-only condition C trains on the imbalanced patches but weights the loss so that every class has the same influence. Neither the balanced reference B nor condition R, which combines both factors, requires reweighting.

Condition	Patch allocation	Class shares in the loss	Varies
B: balanced reference	$r1$	uniform	neither
F: frequency only	$r1$	realized $r100$ shares	frequency
C: coverage only	$r100$	uniform	coverage
R: both	$r100$	realized $r100$ shares	both

Table 1: The four matched conditions. Each condition trains its own low-rank adaptation (LoRA) adapters and linear head from the pretrained encoder. The target $r100$ shares are the class shares realized by the block's imbalanced allocation, not an idealized profile.

Loss weights. Let class c have n_c of the N training patches, and let q_c denote its target share of the loss. Every patch of class c receives the fixed weight

$$w_c = \frac{q_c}{n_c/N}, \tag{1}$$

and the loss of a batch of size b is $\frac{1}{b}\sum_i w_{y_i}\,\ell_i$, where ℓ_i is the cross-entropy of patch i. The weights average to one over the training patches; B and R have unit weights and reduce to ordinary cross-entropy. Dividing by batch size preserves the fixed per-patch weights instead of renormalizing them by each batch's weight sum. The loss and gradient magnitudes can still vary with batch composition, including whether a batch contains rare-class patches.

Pairing. Every condition samples patches uniformly with replacement from its own allocation. B and F draw the same sequence of patch positions, and so do C and R. Changing the class shares in the loss therefore does not change which patches the model sees. Within a block, all four conditions also start from the same adapter and head weights: a random seed derived from the split and draw resets every random number generator before each fit. Different blocks use different seeds. Each fit records digests of its initial weights and sampled positions. Before pooling results, a technical check confirms identical initial weights across all four conditions and identical patch sequences within B/F and C/R pairs.

Training. The recipe follows the mitigation studies: LoRA on every attention projection with rank 8, scaling 16, and no dropout; a linear head; batch size 256; AdamW with learning rate 10^{-4} and weight decay 10^{-4}; a cosine schedule; and 600 (BRACS) or 2,400 (TCGA-UT) updates. Images are decoded deterministically without augmentation. All four conditions use this fixed recipe without condition-specific tuning. Training one block per split first checks numerical and technical soundness, without using effect size or direction as a criterion.

3 Estimands

Let B, F, C, and R denote the patient-macro balanced accuracies of the four conditions, pooled over the 30 blocks of a dataset and expressed as percentages. Damage is positive when imbalance lowers accuracy; all contrasts are in percentage points:

$$D = B - R \qquad \text{total damage,} \tag{2}$$

$$D_F = B - F \qquad \text{frequency-only damage,} \tag{3}$$

$$D_C = B - C \qquad \text{coverage-only damage,} \tag{4}$$

$$I = D - D_F - D_C \qquad \text{interaction.} \tag{5}$$

A positive interaction means that the two factors together cause more damage than the sum of their separate effects. Restoring balanced class influence on the imbalanced patches moves the model from R to C. The resulting recovery and its share of total damage are

$$G = C - R = D_F + I, \qquad h = \frac{G}{D} = 1 - \frac{D_C}{D}. \tag{6}$$

The absolute contrasts are the primary results; h is secondary and is reported with its paired interval, without truncation to $[0, 1]$. A numerical share is interpreted only if the point estimate of D is at least one percentage point and its 95% interval lies strictly above zero; otherwise only the absolute contrasts are interpreted. Replicates with a non-positive denominator are counted and reported, never discarded. Each replicate uses its own denominator rather than the observed estimate.

4 Inference

Uncertainty estimation follows the mitigation studies. The bootstrap combines Bayesian weights on test patients (Rubin, 1981) with resampling of the ten draws within each split and gives each split equal weight. Patient weights are shared across the three splits: within each replicate, a patient who appears in more than one test split receives the same weight in each. All four conditions share the patient weights and resampled draws, so every contrast uses paired replicates. Intervals are 95% percentile intervals from 10,000 replicates; the observed estimate is not included among the replicates used to calculate them. These are marginal estimation intervals for each quantity, not a simultaneous test of all components.

Two diagnostics show where the uncertainty comes from. The patient-only interval keeps every draw at equal weight and resamples only patients; the draw-only interval keeps the observed patients and resamples only draws. Neither replaces the full interval, and their widths do not add. Because each draw has one random seed, the draw-only spread includes initialization variation as well as allocation variation. Per-split estimates show whether a finding depends on one patient split.

5 Interpretation

The experiment yields a matched decomposition for one learner and one training recipe. G is the recovery from restoring balanced class influence when nothing else changes. It is not an

upper bound for augmentation, other losses, or better optimization, and D_C is not an irreducible information limit. The decomposition supports a conditional statement: a method that removes the frequency-related damage and leaves the coverage-only residual unchanged should recover close to G, or a share close to h. Comparisons with the mitigation results are descriptive and distinguish a method's recovery from its recovery after subtracting its gain on balanced data.

6 Results

All 240 fits completed and passed the technical checks. Within every block, the four conditions started from identical weights, B/F and C/R pairs drew identical patch sequences, and the loss targets matched the paired imbalanced allocation. The balanced reference B and condition R reproduce the cross-entropy runs of the mitigation studies: B is 36.49 on BRACS and 71.45 on TCGA-UT (earlier 36.53 and 71.51), and total damage is 5.07 and 1.58 percentage points (earlier 4.95 and 1.56). Table 2 lists all accuracies and contrasts; Figure 1 shows the contrasts with their intervals.

Quantity	BRACS	TCGA-UT
Balanced reference B	36.49 [33.78, 39.26]	71.45 [70.24, 72.65]
Frequency only F	35.24 [32.76, 37.73]	71.25 [70.01, 72.45]
Coverage only C	33.40 [31.01, 35.69]	71.47 [70.28, 72.63]
Both R	31.42 [29.07, 33.71]	69.87 [68.69, 71.03]
Total damage D	5.07 [3.32, 6.98]	1.58 [1.21, 1.94]
Frequency damage D_F	1.26 [-0.13, 2.64]	0.20 [-0.10, 0.50]
Coverage damage D_C	3.09 [1.62, 4.67]	-0.02 [-0.30, 0.25]
Interaction I	0.72 [-0.58, 2.03]	1.40 [1.07, 1.73]
Recovery $G = C - R$	1.97 [1.12, 2.87]	1.60 [1.29, 1.91]
Share $h = G/D$	0.39 [0.24, 0.56]	1.01 [0.86, 1.22]

Table 2: Coverage-only damage accounts for most estimated BRACS damage; the interaction accounts for most on TCGA-UT, where both single-factor intervals span zero. Rows one to four are patient-macro balanced accuracies (%); the remaining rows are contrasts in percentage points, with positive damage meaning lower accuracy under imbalance. Brackets give 95% paired bootstrap intervals.

BRACS. Training on the imbalanced patches with balanced class influence costs 3.09 [1.62, 4.67] points, about three-fifths of the total damage of 5.07 [3.32, 6.98]. Skewing class influence on the balanced patches costs 1.26 [−0.13, 2.64] points, and the interaction is 0.72 [−0.58, 2.03]; neither interval excludes zero. Restoring balanced class influence on the imbalanced patches recovers $G = 1.97$ [1.12, 2.87] points, a share $h = 0.39$ [0.24, 0.56] of total damage. The share meets the interpretation criteria: total damage exceeds one point and its interval lies above zero. No bootstrap replicate has a non-positive denominator. Across the three splits, coverage-only damage ranges from 2.69 to 3.77 points and the share from 0.36 to 0.42.

TCGA-UT. Both single-factor damage estimates are close to zero and their intervals span zero: frequency-only damage is 0.20 [−0.10, 0.50] and coverage-only damage is −0.02 [−0.30, 0.25] points. Together they cost 1.58 [1.21, 1.94] points, with the interaction of 1.40 [1.07, 1.73] accounting for most of the estimated damage. Restoring balanced class influence recovers $G =$ 1.60 [1.29, 1.91] points, a share $h = 1.01$ [0.86, 1.22] of the damage. The share interval includes full recovery but also allows incomplete recovery or recovery beyond the balanced reference.

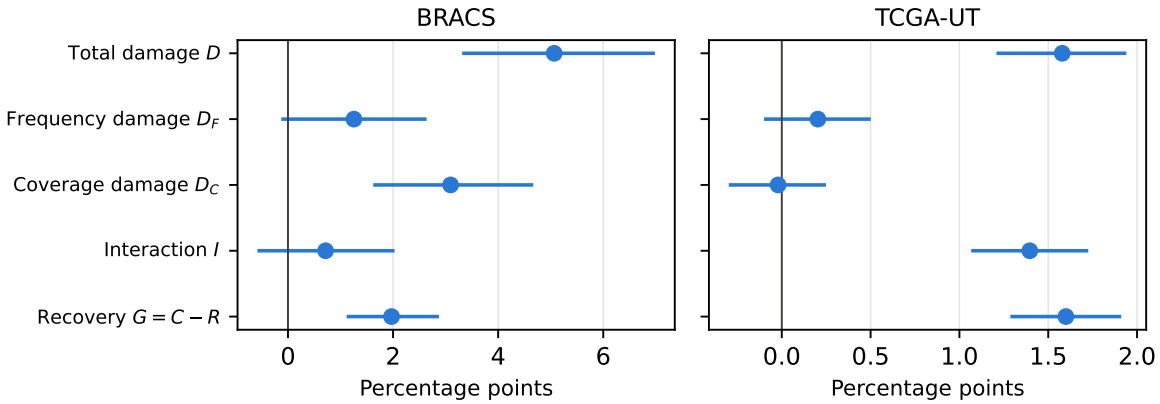


Figure 1: Damage components at $\rho = 100$ under low-rank adaptation (LoRA), with 95% paired intervals. On BRACS, coverage-only damage is the only resolved component besides the total; on TCGA-UT, the interaction is resolved and both single-factor damages are close to zero. The horizontal scales differ between panels.

The coverage-only interval spans small gains and losses rather than establishing exact equality. Across the three splits, the interaction ranges from 1.30 to 1.55 points and the share from 0.95 to 1.08.

Sources of uncertainty. Resampling only patients or only draws gives intervals of similar width, both narrower than the full intervals. On BRACS, for example, frequency-only damage has a patient-only interval of [0.62, 1.94] and a draw-only interval of [0.15, 2.34]; on TCGA-UT, the interaction has corresponding intervals of [1.23, 1.56] and [1.15, 1.63]. Uncertainty therefore comes from both the finite test cohort and variation between allocation draws, which here also includes random initialization.

Comparison with the frozen readout. The decomposition differs from that of the frozen-encoder logistic-regression readout on the same patients (Queisler, 2026a,b), mainly in frequency-only damage. On BRACS, the frozen readout lost 4.47 points to frequency and 3.29 to coverage; under LoRA, frequency-only damage falls to 1.26, while coverage-only damage remains similar at 3.09. On TCGA-UT, the frozen readout lost 1.20 points to frequency, 0.53 to coverage, and 1.18 to the interaction; under LoRA, both single-factor damage estimates are close to zero, while an interaction of similar size, 1.40, remains. The two learners also differ in loss, optimizer, and training length, so this comparison does not isolate an effect of LoRA.

Comparison with the mitigation methods. Figure 2 compares recoveries from the mitigation studies with the benchmark G. These recoveries use the earlier runs and their own estimate of total damage, so the comparison is descriptive. On BRACS, training-time and post-hoc logit adjustment recover 1.96 and 1.94 points, close to $G = 1.97$. Curriculum of data augmentation (CUDA) recovers more, 2.59 points, but subtracting its gain on balanced data leaves only 1.20 points, below the benchmark in point estimate. On TCGA-UT, training-time and post-hoc logit adjustment recover 1.54 and 1.41 points, within the interval for $G = 1.60$. The estimates are descriptively similar to the benchmark; this comparison does not establish equal recovery.

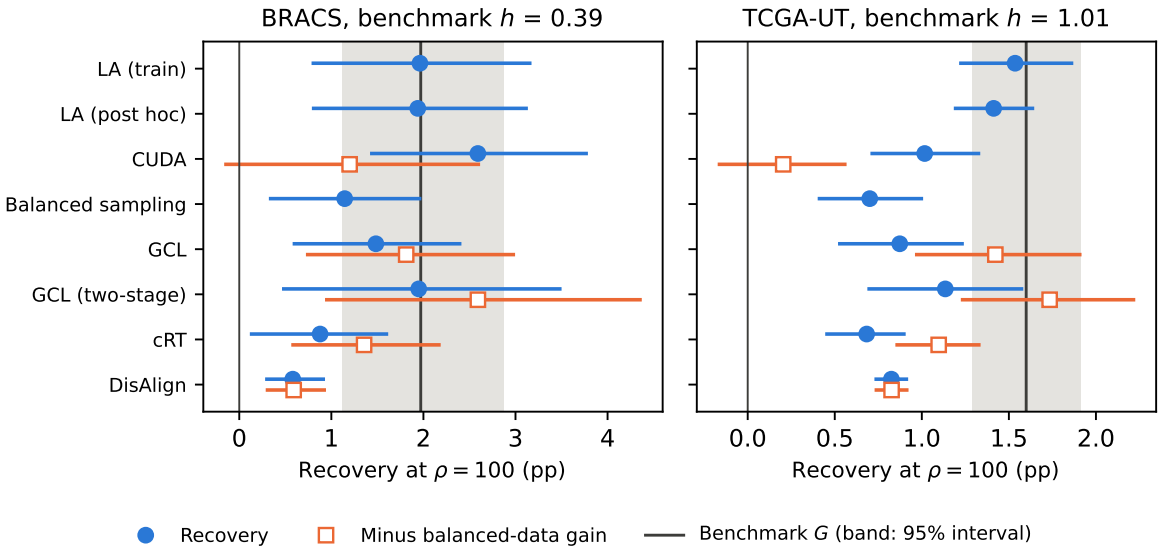


Figure 2: Logit adjustment gives descriptively similar recovery to restoring balanced class influence on both datasets. Points show each mitigation method's recovery at $\rho = 100$ from the mitigation studies (filled) and, where defined, the recovery after subtracting its gain on balanced data (open). The vertical line and band show this experiment's benchmark G and its 95% interval; the panel titles give the share $h = G/D$. Method abbreviations: LA, logit adjustment; CUDA, curriculum of data augmentation; GCL, Gaussian clouded logit adjustment; cRT, classifier re-training.

7 Discussion

Under the LoRA recipe of the mitigation studies, restoring balanced class influence recovers about two-fifths of the BRACS damage and leaves about three points unrecovered. On TCGA-UT, both single-factor damage estimates are close to zero and the interaction accounts for most estimated damage. The coverage-only interval spans small gains and losses, consistent with full recovery.

Logit adjustment recovered close to G: about two-fifths of the BRACS damage and nearly all of the TCGA-UT damage. This fits the conditional statement of Section 5: a method that removes frequency-related damage while leaving the coverage-only residual unchanged should recover close to G. The agreement is descriptive, follows already known mitigation results, and does not establish a shared mechanism with loss reweighting. The BRACS residual concerns uniform loss reweighting under this fixed recipe; whether another weighting scheme, objective, or training recipe could reduce it remains untested by the decomposition.

The comparison with the frozen readout cautions against transferring earlier shares. Frequency-only damage was the largest BRACS component under the frozen readout; coverage-only damage is largest under LoRA. A share estimated for one learner therefore does not carry over to another learner on the same data, motivating the repeated decomposition.

8 Limitations

The decomposition reuses the patients, splits, and allocations inspected in the earlier studies; it does not provide an independent test set. It covers one encoder, one imbalance ratio, one adaptation recipe, and fixed training budgets, so it does not establish how the components behave under other ratios or recipes. Each allocation draw uses one random seed, so initialization and allocation variation cannot be separated. Fits are reproducible only on fixed software

and hardware. Conditions within a block sometimes ran on different graphics processing unit (GPU) models (A100, H100, H200, and, for four TCGA-UT fits, an RTX PRO 6000). They share their initialization, but reduced-precision arithmetic differs slightly between devices and adds a small amount of unpaired training noise to the contrasts. The balanced reference B and condition R reproduce the cross-entropy conditions of the mitigation studies under fresh, matched initialization; the earlier runs serve only as a descriptive reference.

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